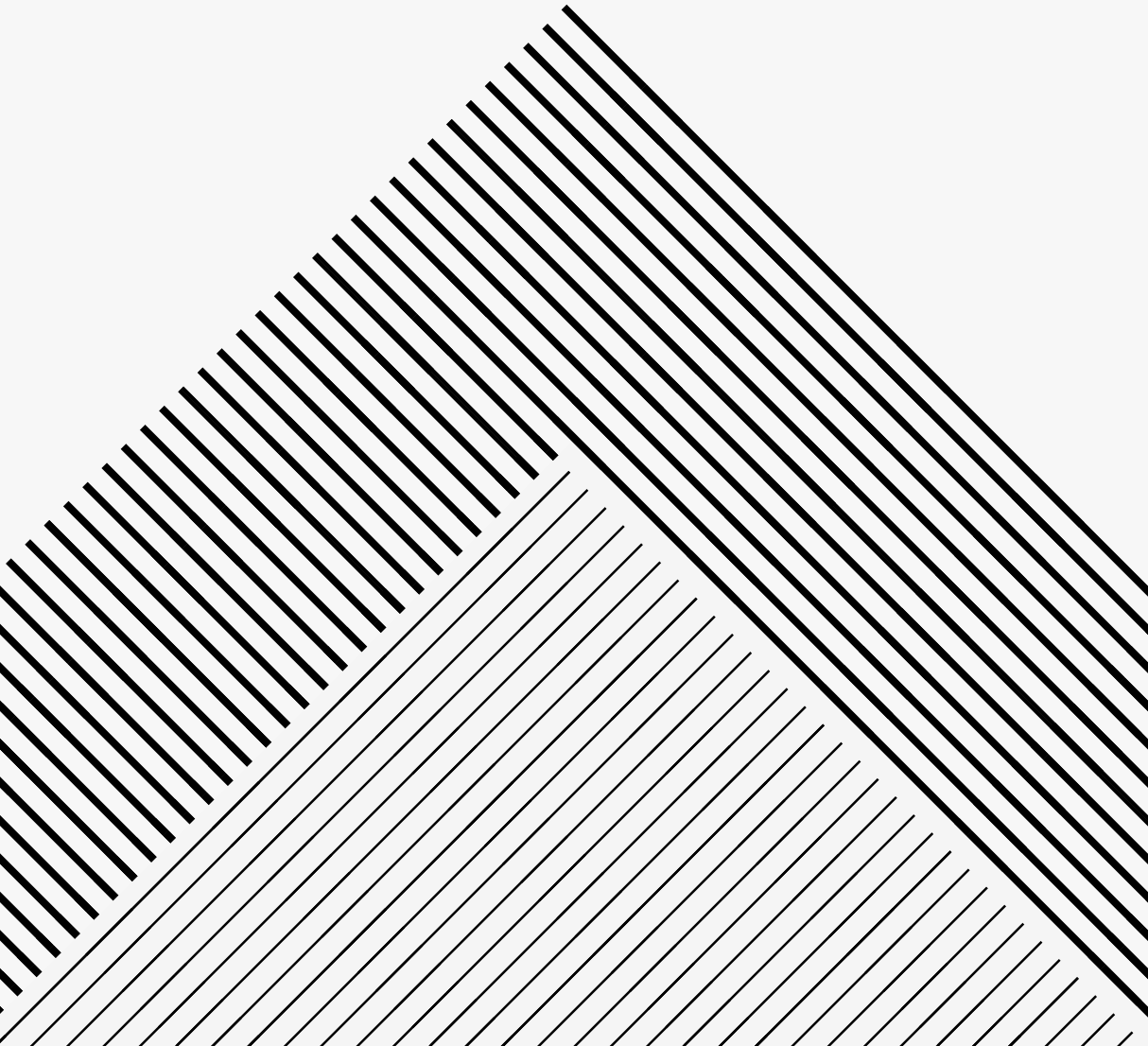
ANALISI MATEMATICA 1

L 9

09-10-26



Asintotici e o-piccoli

Def: Date a_n, b_n successioni ($b_n \neq 0$ definitivamente in $n \in \mathbb{N}$)

- se $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ scriviamo $a_n = o(b_n)$
- se $\lim_{n\to\infty} \frac{a_n}{b_n} = 1$ scriviamo $a_n \sim b_n$

OSS: $a_n \sim b_n \iff a_n = b_n + o(b_n)$

infatti

$$a_n \sim b_n \iff \lim_{n\to\infty} \frac{a_n}{b_n} = 1$$

$$\iff \lim_{n\to\infty} \left(\frac{a_n}{b_n} - 1\right) = 0$$

$$\iff \lim_{n\to\infty} \frac{a_n - b_n}{b_n} = 0$$

$$\iff a_n - b_n = o(b_n)$$

$$\iff a_n = (a_n - b_n) + b_n$$

$$= b_n + o(b_n)$$

OSS: $b_n = o(1)$ per definizione

$$\lim_{n\to\infty} \frac{b_n}{1} = 0$$

cioè $\lim_{n\to\infty} b_n = 0$, cioè b_n è infinitesima

Oss:

- $a_n \sim a_n$ ovvio
- $a_n \sim b_n$ e $b_n \sim c_n \Rightarrow a_n \sim c_n$

  Infatti

$$\frac{a_n}{c_n} = \frac{a_n}{b_n}\,\frac{b_n}{c_n} \longrightarrow 1 \cdot 1 = 1$$ ✓

  ($\frac{a_n}{b_n} \to 1$, $\frac{b_n}{c_n} \to 1$)

- $a_n \sim b_n \Rightarrow b_n \sim a_n$

  Infatti

$$\frac{b_n}{a_n} = \left(\frac{a_n}{b_n}\right)^{-1} \longrightarrow 1^{-1} = 1$$

  ($\frac{a_n}{b_n} \to 1$)

$\sim$ è una relazione di equivalenza fra successioni (definitivamente $\neq 0$)

ES: FOGLIO ASINTOTICI NOTEVOLI SU WEBEEP

ES: Per $n \to \infty$

- $\sin \frac{1}{n} \sim \frac{1}{n}$

$$\sin \frac{1}{n} = \frac{1}{n} + o\left(\frac{1}{n}\right)$$

- $e^{\frac{1}{n}} - 1 \sim \frac{1}{n}$

$$e^{\frac{1}{n}} - 1 = \frac{1}{n} + o\left(\frac{1}{n}\right)$$

$$\Rightarrow \quad e^{\frac{1}{n}} = 1 + \frac{1}{n} + o\left(\frac{1}{n}\right)$$

- $\log\left(1+\frac{1}{n}\right) \sim \frac{1}{n}$

$$\log\left(1+\frac{1}{n}\right) = \frac{1}{n} + o\left(\frac{1}{n}\right)$$

- $1 - \cos \frac{1}{n} \sim \frac{1}{2n^2}$

$$1 - \cos \frac{1}{n} = \frac{1}{2n^2} + o\left(\frac{1}{n^2}\right)$$

$$\Rightarrow \quad \cos \frac{1}{n} = 1 - \frac{1}{2n^2} + o\left(\frac{1}{n^2}\right)$$

- $\left(1+\frac{1}{n}\right)^{\alpha} - 1 \sim \frac{\alpha}{n}$ $\qquad \alpha \neq 0$

$$\left(1+\frac{1}{n}\right)^{\alpha} - 1 = \frac{\alpha}{n} + o\left(\frac{1}{n}\right)$$

$$\Rightarrow \left(1+\frac{1}{n}\right)^{\alpha} = 1 + \frac{\alpha}{n} + o\left(\frac{1}{n}\right)$$

- $\arctan \frac{1}{n} \sim \frac{1}{n}$

  $\arctan \frac{1}{n} = \frac{1}{n} + o\left(\frac{1}{n}\right)$

- ...

Prop: 1) Se $a_n \sim a'_n$ e $b_n \sim b'_n$

$\Rightarrow a_n b_n \sim a'_n \cdot b'_n$

2) Se $a_n \sim a'_n$ e $b_n \sim b'_n$

$\Rightarrow \frac{a_n}{b_n} \sim \frac{a'_n}{b'_n}$

3) Se $a_n \sim a'_n$, $\alpha \in \mathbb{R}$

$\Rightarrow a_n^{\alpha} \sim (a'_n)^{\alpha}$

(assumendo che l'operazione abbia senso...)

4) Se $a_n \sim a'_n$ allora le due successioni hanno lo stesso carattere, cioè

o convergono entrambe allo stesso limite $l \in \mathbb{R}$ o divergono entrambe a $+\infty$ o entrambe a $-\infty$, o sono entrambe indeterminate

OSS: In generale $a_n \sim a'_n$, $b_n \sim b'_n$

$\not\Rightarrow$ $a_n \pm b_n \sim a'_n \pm b'_n$

$a_n^{b_n} \sim (a'_n)^{b'_n}$

$e^{a_n} \sim e^{a'_n}$

$\log(a_n) \sim \log(a'_n)$

OSS: In generale

$a_n \sim b_n \not\Rightarrow a_n - b_n \to 0$

$a_n - b_n \to 0 \not\Rightarrow a_n \sim b_n$

dim (della proposizione)

$a_n \sim a'_n$, $b_n \sim b'_n$

1) $\Rightarrow a_n b_n \sim a'_n b'_n$

$$\frac{a'_n b'_n}{a_n b_n} = \left(\frac{a'_n}{a_n}\right)^{\to 1} \cdot \left(\frac{b'_n}{b_n}\right)^{\to 1} \to 1 \cdot 1 = 1$$

2) $\dots \Rightarrow \dfrac{a_n}{b_n} \sim \dfrac{a'_n}{b'_n}$

$$\frac{\frac{a_n}{b_n}}{\frac{a'_n}{b'_n}} = \frac{a_n}{b_n} \cdot \frac{b'_n}{a'_n} = \left(\frac{a_n}{a'_n}\right) \cdot \left(\frac{b_n}{b'_n}\right)^{-1}$$

$\frac{a_n}{a'_n} \to 1$, $\frac{b_n}{b'_n} \to 1$

$\longrightarrow 1 \cdot 1^{-1} = 1$

3) $\dots \Rightarrow (a_n)^\alpha \sim (a'_n)^\alpha$

$$\frac{a_n^\alpha}{(a'_n)^\alpha} = \left(\frac{a_n}{a'_n}\right)^\alpha \longrightarrow 1^\alpha = 1$$

$\frac{a_n}{a'_n} \to 1$

4) $\dots \Rightarrow a_n, a'_n$ hanno lo stesso comportamento al limite

Se $\exists \lim_{n\to\infty} a'_n = \ell \in \overline{\mathbb{R}}$

$$a_n = \frac{a_n}{a'_n} \cdot a'_n \longrightarrow 1 \cdot \ell = \ell \in \overline{\mathbb{R}}$$

$\frac{a_n}{a'_n} \to 1$, $a'_n \to \ell$

Se invece a'_n non ammette limite

in $\overline{\mathbb{R}}$ e per assurdo $\exists \lim_{n\to\infty} a_n = \ell \in \overline{\mathbb{R}}$

allora

$$a'_n = \frac{a'_n}{a_n} \cdot a_n \longrightarrow 1 \cdot \ell = \ell \in \overline{\mathbb{R}}$$

cioè $\exists \lim_{n\to\infty} a'_n = \ell \in \overline{\mathbb{R}}$, assurdo

$\Rightarrow \nexists \lim_{n\to\infty} a_n$ in $\overline{\mathbb{R}}$ □

Es: $$a_n = \frac{3n^7 + 5n^{42/95} + \log n - \cos n + 5}{2n^6 - 18n^3 + \arctan n + (\log n)^2 - 2}$$

$$N = 3n^7 + o(n^7) \sim 3n^7$$

$$D = 2n^6 + o(n^6) \sim 2n^6$$

$$a_n = \frac{3n^7 + o(n^7)}{2n^6 + o(n^6)} \sim \frac{3n^7}{2n^6} = \frac{3}{2} n \longrightarrow +\infty$$

$$\Rightarrow \lim_{n\to\infty} a_n = +\infty$$

Es: (fogli asintotici su WeBeep)

Se $\varepsilon_n \to 0$ ($\varepsilon_n \neq 0$ definitivamente) allora

- $\sin \varepsilon_n \sim \varepsilon_n$ $\quad \sin \varepsilon_n = \varepsilon_n + o(\varepsilon_n)$

- $1-\cos \varepsilon_n \sim \frac{\varepsilon_n^2}{2}$ $\quad 1-\cos \varepsilon_n = \frac{\varepsilon_n^2}{2} + o(\varepsilon_n^2)$

$$\cos \varepsilon_n = 1 - \frac{\varepsilon_n^2}{2} + o(\varepsilon_n^2)$$

- $e^{\varepsilon_n} - 1 \sim \varepsilon_n$ $\quad e^{\varepsilon_n} - 1 = \varepsilon_n + o(\varepsilon_n)$

$$e^{\varepsilon_n} = 1 + \varepsilon_n + o(\varepsilon_n)$$

- $\log(1+\varepsilon_n) \sim \varepsilon_n$ $\quad \log(1+\varepsilon_n) = \varepsilon_n + o(\varepsilon_n)$

- $(1+\varepsilon_n)^\alpha - 1 \sim \alpha \varepsilon_n$ $\quad (1+\varepsilon_n)^\alpha - 1 = \alpha \varepsilon_n + o(\varepsilon_n)$

$$(1+\varepsilon_n)^\alpha = 1 + \alpha \varepsilon_n + o(\varepsilon_n)$$

$(\alpha \neq 0)$

...

ES: $a_n = \dfrac{\sin \frac{1}{n^2}}{e^{7/n} - 1} \cdot n$

$\varepsilon_n = \frac{1}{n^2} \to 0$ $\quad \sin \varepsilon_n \sim \varepsilon_n$

$$\sin\frac{1}{n^2} \sim \frac{1}{n^2}$$

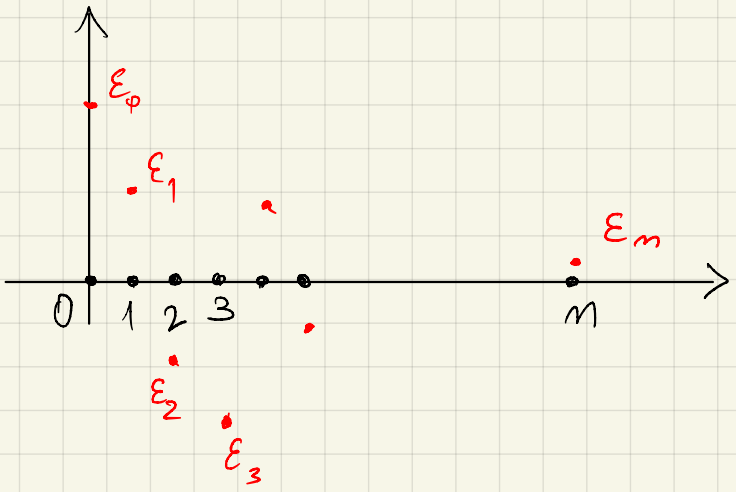
$\varepsilon_n = \frac{7}{n} \to 0$ $\qquad e^{\varepsilon_n} - 1 \sim \varepsilon_n$

$$e^{7/n} - 1 \sim 7/n$$

$$a_n \sim \frac{\frac{1}{n^2}}{\frac{7}{n}} \cdot n = \frac{1}{7}$$

$$\Rightarrow \lim_{n\to\infty} a_n = \frac{1}{7}$$

OSS:



ES:
$$a_n = \frac{\sqrt[3]{1+n} - n^{1/3}}{1 - e^{2/n}} \cdot \arctan\frac{1}{n^2} \cdot n^{5/3}$$

- $\arctan\frac{1}{n^2} \sim \frac{1}{n^2}$

$\arctan\varepsilon_n \sim \varepsilon_n$ $\qquad \varepsilon_n = \frac{1}{n^2} \to 0$

- $1 - e^{2/n} \sim -\frac{2}{n}$

$+ \left(e^{\varepsilon_n} - 1\right) \sim + \varepsilon_n \qquad \varepsilon_n = \frac{2}{n} \to 0$

- $\sqrt[3]{1+n} - n^{1/3} \sim \; ?$

$\left(1+\varepsilon_n\right)^\alpha - 1 \sim \alpha \varepsilon_n \qquad \varepsilon_n \to 0$

$\left(1+\varepsilon_n\right)^\alpha = 1 + \alpha \varepsilon_n + o(\varepsilon_n)$

[<u>No</u>: • $\sqrt[3]{1+n} = \left(1+n\right)^{1/3} \neq 1 + \frac{n}{3} + o(n)$

perché $\varepsilon_n = n \not\to 0$

• $\sqrt[3]{1+n} \sim n^{1/3}$ vero

$\not\Rightarrow \sqrt[3]{1+n} - n^{1/3} \sim n^{1/3} - n^{1/3} = 0$

no ~ in somme e sottrazioni]

$\sqrt[3]{1+n} = n^{1/3} \underbrace{\left(1 + \frac{1}{n}\right)^{1/3}}_{(1+\varepsilon_n)^\alpha \qquad \varepsilon_n = \frac{1}{n} \quad \alpha = \frac{1}{3}}$

$= n^{1/3} \left(1 + \frac{1}{3n} + o\left(\frac{1}{n}\right)\right)$

$= n^{1/3} + \frac{1}{3n^{2/3}} + \underbrace{n^{1/3} \cdot o\left(\frac{1}{n}\right)}_{o\left(\frac{1}{n^{2/3}}\right)}$

Infatti

$$\frac{n^{1/3}\cdot o\left(\frac{1}{n}\right)}{\frac{1}{n^{2/3}}} = \frac{o\left(\frac{1}{n}\right)}{\frac{1}{n}} \to 0$$

$$\sqrt[3]{1+n} - n^{1/3} = n^{1/3} + \frac{1}{3n^{2/3}} + o\left(\frac{1}{n^{2/3}}\right) - n^{1/3}$$

$$= \frac{1}{3n^{2/3}} + o\left(\frac{1}{n^{2/3}}\right)$$

$$\sim \frac{1}{3n^{2/3}}$$

$$a_n = \ldots \sim \frac{\frac{1}{3n^{2/3}}}{-\frac{2}{n}} \cdot \frac{1}{n^2} \cdot n^{5/3} = \frac{-1}{6}$$

ES: $a_n = (n^3+n^2)^{1/3} - (n^3-n)^{1/3}$

$$a^3 - b^3 = (a-b)(a^2+ab+b^2)$$

$$a = \sqrt[3]{n^3+n^2} \qquad b = \sqrt[3]{n^3-n}$$

$$a-b = \frac{a^3-b^3}{a^2+ab+b^2} \quad \ldots$$

$$(n^3+n^2)^{1/3} = n\left(1+\frac{1}{n}\right)^{1/3} \qquad \sim n$$

$$= n\left(1+\frac{1}{3n}+o\left(\frac{1}{n}\right)\right)$$

$$(n^3-n)^{1/3} = n\left(1-\frac{1}{n^2}\right)^{1/3} \qquad \sim n$$

$$= n\left(1-\frac{1}{3n^2}+o\left(\frac{1}{n^2}\right)\right)$$

$$(1+\varepsilon_n)^\alpha = 1+\alpha\varepsilon_n+o(\varepsilon_n)$$

$$a_n = (n^3+n^2)^{1/3}-(n^3-n)^{1/3}$$

$$= n\left(1+\frac{1}{3n}+o\left(\frac{1}{n}\right)\right)-n\left(1-\frac{1}{3n^2}+o\left(\frac{1}{n^2}\right)\right)$$

$$= \cancel{n}+\frac{1}{3}+n\,o\left(\frac{1}{n}\right)-\cancel{n}+\frac{1}{3n}+n\,o\left(\frac{1}{n^2}\right)$$

$$= \frac{1}{3}+\frac{1}{3n}+\underbrace{n\,o\left(\frac{1}{n}\right)}_{o(1)}+\underbrace{n\,o\left(\frac{1}{n^2}\right)}_{o\left(\frac{1}{n}\right)}$$

$$o(1)$$

infatti:

$$\frac{n\cdot o\left(\frac{1}{n}\right)}{1} = \frac{o\left(\frac{1}{n}\right)}{\frac{1}{n}} \longrightarrow 0$$

$$\frac{n \cdot o\left(\frac{1}{n^2}\right)}{\frac{1}{n}} = \frac{o\left(\frac{1}{n^2}\right)}{\frac{1}{n^2}} \longrightarrow 0$$

$$a_n = \frac{1}{3} + \frac{1}{3n} + o(1)$$

$$= \frac{1}{3} + o(1) \longrightarrow \frac{1}{3}$$

OSS:

- $a_n \cdot o(b_n) = o(a_n b_n)$

  Infatti

  $$\frac{a_n \cdot o(b_n)}{a_n b_n} = \frac{o(b_n)}{b_n} \longrightarrow 0$$

  per definizione

- $o(a_n) = a_n \cdot o(1)$
- $c\,o(a_n) = o(a_n) \quad \forall c \neq 0$

  Infatti

  $$\frac{c\,o(a_n)}{a_n} \longrightarrow c \cdot 0 = 0$$

- $\pm o(a_n) = o(a_n)$
- $o\left(\frac{1}{n}\right) + o\left(\frac{1}{n^2}\right) = o\left(\frac{1}{n}\right)$
- $o(n) + o(n^2) = o(n^2)$
- se $a_n \sim b_n \Rightarrow o(a_n) = o(b_n)$

  infatti

$$\frac{o(a_n)}{b_n} = \frac{o(a_n)}{a_n} \cdot \frac{a_n}{b_n} \to 0$$

$\frac{o(a_n)}{a_n} \to 0$, $\frac{a_n}{b_n} \to 1$

ad esempio

$$o\left(\sin\frac{1}{n}\right) = o\left(\frac{1}{n}\right)$$

Es: $$a_n = \frac{\sin\frac{1}{n} - \frac{1}{n}}{\frac{1}{n^3}}$$

$\left[\frac{0}{0}\right]$ forma indeterminata

$\sin \varepsilon_n \sim \varepsilon_n$ $\qquad \varepsilon_n = \frac{1}{n} \to 0$

$\sin \varepsilon_n = \varepsilon_n + o(\varepsilon_n)$

$$\sin\frac{1}{n} - \frac{1}{n} = \frac{1}{n} + \sigma\left(\frac{1}{n}\right) - \frac{1}{n} = \sigma\left(\frac{1}{n}\right)$$

$$\Rightarrow \quad a_n = \frac{\sigma\left(\frac{1}{n}\right)}{\frac{1}{n^3}} = n^3\,\sigma\left(\frac{1}{n}\right) = \sigma(n^2)$$

$$\lim_{n\to\infty} a_n = ?$$